MACHINE LEARNING / MEDICAL IMAGE BENCHMARK

Blood cell image classification

Comparing a Random Forest baseline with a compact convolutional neural network.

Course project | Cleaned edition of the original BloodMNIST notebooks

8 cell classes	11,959 train	3,421 test
28 x 28 RGB images	Official training split	Official test split

Research question

Can a small convolutional network improve on a pixel-based Random Forest for classifying eight types of blood cells? The project compares overall discrimination and class-level performance on BloodMNIST.

Model design

Random Forest	Convolutional neural network
Flattened RGB inputs in [0, 1]; 100 trees; random_state=42.	Two convolution/pooling blocks (32 and 64 channels), a 128-unit hidden layer and eight logits.
Classification from raw pixel features.	Adam, learning rate 0.001; batch 128; ten epochs; unweighted cross-entropy; RGB normalization.

What the saved run showed

The CNN achieved 90.76% test accuracy versus 83.10% for the Random Forest, a difference of 7.66 percentage points. Macro F1 increased from 0.8004 to 0.8923. No uncertainty interval or external validation was available.

Provenance: values come from saved outputs in the original notebooks. Cleaned notebooks have cleared outputs and have not been retrained in this preparation environment.

Better overall scores; uneven class performance

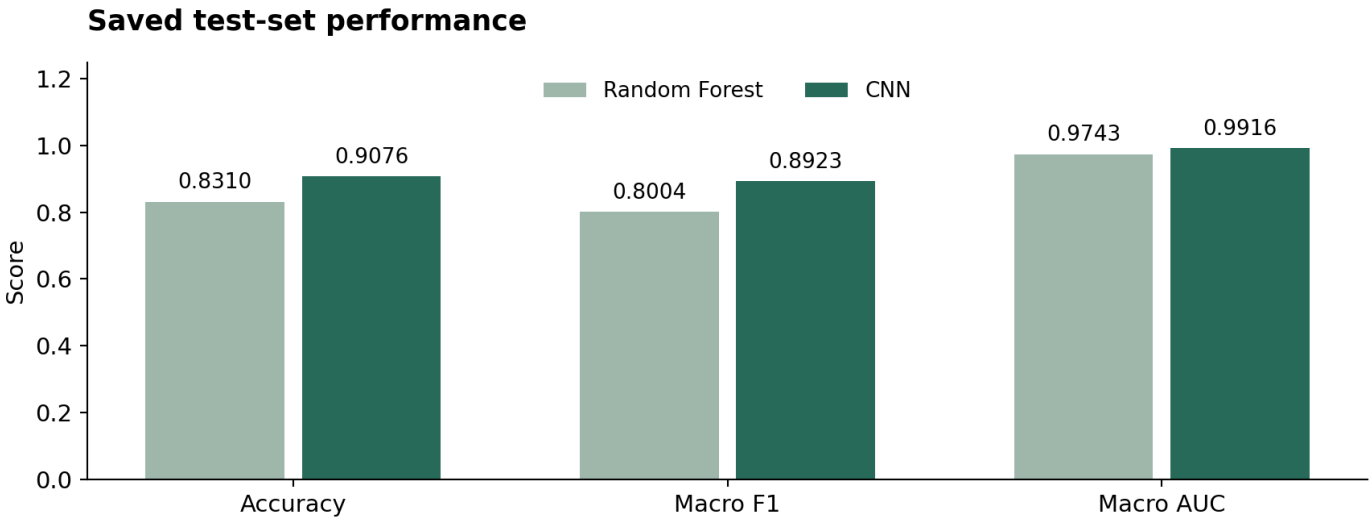


Figure 1. Archived test metrics. AUC is the macro one-vs-rest score; bars are point estimates, without uncertainty intervals.

Model	Accuracy	Macro F1	Macro AUC
Random Forest	83.10%	0.8004	0.9743
CNN	90.76%	0.8923	0.9916

Class-level errors

CNN F1 was 0.9989 for platelets and 0.9809 for eosinophils, compared with 0.7836 for monocytes and 0.7982 for immature granulocytes. These differences motivate class-level evaluation rather than reliance on accuracy alone.

Reproducibility and next steps

The repository retains the original notebooks and adds clean entry points, dependency instructions and separate CSV exports for future runs. A CNN seed was added during cleanup, so exact recovery of the old stochastic run is not guaranteed. Use the official validation split for future tuning and keep the test split reserved for final evaluation.

Source: github.com/jengithubb/DermaMNIST-CNN-project, commit 243db78. Despite its old name, the source uses BloodMNIST. The README's weighted-loss claim and 90.1% score were corrected to match the training cell and saved output. Dataset: medmnist.com. This is a benchmark study, not a clinical decision tool.